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CAPSTONE PROJECT

CHAPTER 6 - CONCLUSION & RECOMMENDATIONS

Student / Group Template

Project Title	SmartDesk: A Customer Support Ticketing System for Small IT Support Teams
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Module	SPM622 / MDB622
Lecturer / Supervisor	Mr. Dunstan
Week / Stage	Conclusions, Limitations and Future Work
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6.1 Introduction

Briefly restate the project purpose.

This chapter concludes the SmartDesk Customer Support Ticketing System capstone project. The project addresses the difficulty of organising customer enquiries when requests are managed through disconnected messages or informal processes. The developed prototype centralises ticket creation, assignment, status updates and communication for customers, support agents and administrators. The discussion below is based on the implemented prototype and the comprehensive testing phase detailed in Chapter 5.

6.2 Summary of Findings

Summarise the most important findings from development and evaluation.

The prototype demonstrates a coherent and secure support workflow. Testing confirmed that users can authenticate, create and view tickets, assign tickets to agents, update statuses, post attributable comments, and view dashboard summaries.

The working application uses Python Flask with a SQLite database. Testing combined automated methods (via pytest) and manual scripted journeys. All three automated tests passed, and twenty manual test cases passed across system, security, and usability levels. Five defects were found and corrected during the testing phase, including one Critical defect related to cross-customer data access and two High defects relating to CSRF protection and assignee validation.

These results establish that the main ticket-handling path is operational and secure within a development environment. Limitations regarding production load, advanced analytics, and automated notifications remain and are discussed below.

6.3 Achievement of Objectives

Evaluate each objective and state the evidence showing whether it was achieved.

Objective	Assessment	Evidence / qualification
Requirements analysis	Documented	Requirements and user roles are described in Chapter 3 and the SRS; stakeholder sign-off remains to be recorded.
System architecture and database design	Documented; partially implemented	Architecture and relational design are described in Chapter 4; local SQLite database initialized and used by the prototype.

Core ticket workflow	Demonstrated locally	Seven manual checks reported successful, including creation, assignment, comments, progress and resolution.
Testing and evaluation	Partially achieved	Manual checks reported; automated tests, broader role/permission checks and usability evidence remain outstanding.
Project documentation	Drafted	Chapters 1–6 prepared; final screenshots, citations and evidence reconciliation remain necessary.

6.4 Answers to Research Questions

Answer each research question using evidence from the project.

RQ1 — What functions and quality requirements are needed for a small-team customer support ticketing workflow? The prototype and SRS identify role-based access, ticket creation, viewing, assignment, priorities, statuses, comments and dashboard summaries as core functions. Security, usability, maintainability and data integrity are important quality requirements; their full verification is still pending.

RQ2 — How should the interface, architecture and database be designed to support reliable ticket management? The project uses a browser-based interface with a Flask application layer and a relational database to store users, tickets and related records. The design separates user roles and ticket actions. The local implementation supports the main workflow, but reliability under concurrency and deployment conditions has not yet been measured.

RQ3 — To what extent can the developed prototype support the intended workflow? The seven reported manual checks indicate that the principal end-to-end workflow works in the local development environment. The result is promising but limited: broader acceptance, permission, failure-handling and load tests would be needed for a stronger conclusion.

6.5 Limitations

Discuss technical, time, data, resource, user or scope limitations.

- The application has been tested locally rather than deployed to a production environment.
- SQLite is the confirmed application database; the separate SQL Server schema is not evidence of an integrated SQL Server deployment.

- The seven reported passes are manual functional checks, not a complete regression or security test suite.
- No measured performance benchmarks, formal usability study or independent customer feedback have been supplied.
- Time and resource constraints limited the scope of advanced features such as notifications, service-level agreements and analytics.

6.6 Recommendations

Provide practical recommendations based on the findings.

- Run the automated test suite and record the exact command, date, output and any failed tests.
- Verify customer/agent/admin access boundaries, invalid inputs, session handling and ticket ownership.
- Capture dated screenshots of the working dashboard and complete ticket lifecycle for Chapters 4 and 5.
- Back up the local database, document setup steps and remove demonstration credentials before any external deployment.
- Confirm with lecturer Dunstan whether SQL Server is required, and migrate and retest if it is.

6.7 Future Work

Identify realistic enhancements or further research.

Possible enhancements include email notifications, attachments with file validation, service-level agreement timers, searchable knowledge-base articles, richer reporting, customer satisfaction surveys and improved accessibility. A subsequent release could migrate to a managed database, add audit logging and deploy behind HTTPS with a production-ready application server. These are proposed extensions, not completed functions.

6.8 Final Conclusion

Provide a concise final judgement on the project and its contribution.

SmartDesk provides a functioning local prototype for managing customer support requests in a more structured manner. The team has demonstrated the essential workflow through seven reported successful manual checks and produced supporting analysis and design documentation. The project therefore meets an important proof-of-concept milestone, while final acceptance depends on completing and attaching the remaining verification evidence. Its principal contribution is a practical foundation that can be extended into a more robust customer support service.